

Maven Exercise

Q1) Following parameters are to be considered for exercise submission:

- The complete project has to be submitted. - .idea, target directories should be excluded
- A text file containing the relevant URLs which are referenced and any other documentation.

```
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises$ mvn archetype:generate -DgroupId=com.bootcamp.mavenassignment -DartifactId=maven_assignment -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] Parameter: basedir, Value: /home/ishmeet-singh/IdeaProjects/BootcampExercises
[INFO] Parameter: package, Value: com.bootcamp.mavenassignment
[INFO] Parameter: groupId, Value: com.bootcamp.mavenassignment
[INFO] Parameter: artifactId, Value: maven_assignment
[INFO] Parameter: packageName, Value: com.bootcamp.mavenassignment
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven_assignment
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.904 s
[INFO] Finished at: 2026-02-06T21:18:35+05:30
[INFO]
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises$ cd maven_assignment/
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven_assignment$ ls
pom.xml  src
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven_assignment$
```

Q2) Add a maven dependency and its related repository URL.

```
<?xml version="1.0" encoding="UTF-8" ?>
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.bootcamp.mavenassignment</groupId>
  <artifactId>maven_assignment</artifactId>
  <version>1.0-SNAPSHOT</version>
  <packaging>jar</packaging>
  <name>maven_assignment</name>

  <repositories>
    <repository>
      <id>maven-central</id>
      <url>https://repo.maven.apache.org/maven2</url>
    </repository>
  </repositories>

  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>
    <dependency>
      <groupId>com.mysql</groupId>
      <artifactId>mysql-connector-j</artifactId>
      <version>9.0.0</version>
      <scope>compile</scope>
    </dependency>
  </dependencies>
</project>
```

```
ishmeet-singh@TTNPL-ishmeet: ~/IdeaProjects/BootcampExercises/maven_assignment
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven_assignment$ mvn clean compile
[INFO] Scanning for projects...
[INFO] -----< com.bootcamp.mavenassignment:maven_assignment >-----
[INFO] Building maven_assignment 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
Downloading from maven-central: https://repo.maven.apache.org/maven2/com/mysql/mysql-connector-j/9.6.0/mysql-connector-j-9.6.0.pom
Downloaded from maven-central: https://repo.maven.apache.org/maven2/com/mysql/mysql-connector-j/9.6.0/mysql-connector-j-9.6.0.pom (3.2 kB at 11 kB/s)
Downloading from maven-central: https://repo.maven.apache.org/maven2/com/google/protobuf/protobuf-java/4.31.1/protobuf-java-4.31.1.pom
Downloaded from maven-central: https://repo.maven.apache.org/maven2/com/google/protobuf/protobuf-java/4.31.1/protobuf-java-4.31.1.pom (1.6 kB at 10 kB/s)
Downloading from maven-central: https://repo.maven.apache.org/maven2/com/google/protobuf/protobuf-parent/4.31.1/protobuf-parent-4.31.1.pom
Downloaded from maven-central: https://repo.maven.apache.org/maven2/com/google/protobuf/protobuf-parent/4.31.1/protobuf-parent-4.31.1.pom (9.3 kB at 59 kB/s)
Downloading from maven-central: https://repo.maven.apache.org/maven2/com/google/protobuf/protobuf-bom/4.31.1/protobuf-bom-4.31.1.pom
Downloaded from maven-central: https://repo.maven.apache.org/maven2/com/google/protobuf/protobuf-bom/4.31.1/protobuf-bom-4.31.1.pom (3.7 kB at 34 kB/s)
Downloading from maven-central: https://repo.maven.apache.org/maven2/com/google/protobuf/protobuf-java/4.31.1/protobuf-java-4.31.1.jar
Downloaded from maven-central: https://repo.maven.apache.org/maven2/com/google/protobuf/protobuf-java/4.31.1/protobuf-java-4.31.1.jar (1.9 MB at 3.6 MB/s)
Downloading from maven-central: https://repo.maven.apache.org/maven2/com/mysql/mysql-connector-j/9.6.0/mysql-connector-j-9.6.0.jar
Downloaded from maven-central: https://repo.maven.apache.org/maven2/com/mysql/mysql-connector-j/9.6.0/mysql-connector-j-9.6.0.jar (2.6 MB at 5.0 MB/s)
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ maven_assignment ---
[INFO] Deleting /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven_assignment/target
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ maven_assignment ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven_assignment/src/main/resources
[INFO] --- maven-compiler-plugin:3.11.0:compile (default-compile) @ maven_assignment ---
[INFO] Changes detected - recompiling the module! :dependency
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 21] to target/classes
[INFO] BUILD SUCCESS
[INFO] Total time: 2.198 s
[INFO] Finished at: 2026-02-06T21:38:36+05:30
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven_assignment$
```

Q3) Add a new repository in the pom.xml and use its dependencies.

```
<?xml version="1.0" encoding="UTF-8" ?>
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.bootcamp</groupId>
    <artifactId>maven_assignment</artifactId>
    <version>1.0-SNAPSHOT</version>
    <packaging>jar</packaging>
    <name>maven_assignment</name>
    <description></description>
    <url></url>
    <repositories>
        <repository>
            <id>maven-central</id>
            <url>https://repo.maven.apache.org/maven2</url>
        </repository>
        <repository>
            <id>spring-releases</id>
            <url>https://repo.spring.io/release</url>
        </repository>
    </repositories>
    <dependencies>
        <dependency>
            <groupId>junit</groupId>
            <artifactId>junit</artifactId>
            <version>3.8.1</version>
            <scope>test</scope>
        </dependency>
        <dependency>
            <groupId>com.mysql</groupId>
            <artifactId>mysql-connector-j</artifactId>
            <version>9.6.0</version>
            <scope>compile</scope>
        </dependency>
        <dependency>
            <groupId>org.springframework</groupId>
            <artifactId>spring-core</artifactId>
            <version>6.1.3</version>
        </dependency>
    </dependencies>
</project>
```

```

ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven_assignment$ mvn clean compile
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.bootcamp.mavenassignment:maven_assignment >-----
[INFO] Building maven_assignment 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO] Downloading from maven-central: https://repo.maven.apache.org/maven2/org/springframework/spring-core/6.1.3/spring-core-6.1.3.pom
Downloaded from maven-central: https://repo.maven.apache.org/maven2/org/springframework/spring-core/6.1.3/spring-core-6.1.3.pom (2.0 kB at 5.2 kB/s)
[INFO] Downloading from maven-central: https://repo.maven.apache.org/maven2/org/springframework/spring-jcl/6.1.3/spring-jcl-6.1.3.pom
Downloaded from maven-central: https://repo.maven.apache.org/maven2/org/springframework/spring-jcl/6.1.3/spring-jcl-6.1.3.pom (1.8 kB at 27 kB/s)
[INFO] Downloading from maven-central: https://repo.maven.apache.org/maven2/org/springframework/spring-core/6.1.3/spring-core-6.1.3.jar
Downloaded from maven-central: https://repo.maven.apache.org/maven2/org/springframework/spring-core/6.1.3/spring-core-6.1.3.jar (25 kB at 189 kB/s)
[INFO] Downloading from maven-central: https://repo.maven.apache.org/maven2/org/springframework/spring-jcl/6.1.3/spring-jcl-6.1.3.jar
Downloaded from maven-central: https://repo.maven.apache.org/maven2/org/springframework/spring-jcl/6.1.3/spring-jcl-6.1.3.jar (1.9 MB at 738 kB/s)
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ maven_assignment ---
[INFO] Deleting /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven_assignment/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ maven_assignment ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven_assignment/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:compile (default-compile) @ maven_assignment ---
[INFO] Changes detected - recompiling the module! :dependency
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 21] to target/classes
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 4.023 s
[INFO] Finished at: 2026-02-06T21:43:06+05:30
[INFO]
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven_assignment$

```

Q4) Using JAR plugin, make changes in the pom.xml to make the jar executable. Using `java -jar JAR_NAME`, the output should be printed as "Hello World"

```

<?xml version="1.0" encoding="UTF-8" ?>
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.bootcamp</groupId>
    <artifactId>maven_assignment</artifactId>
    <version>1.0-SNAPSHOT</version>
    <packaging>jar</packaging>
    <name>maven_assignment</name>
    <description></description>
    <url></url>
    <inceptionYear>2026</inceptionYear>
    <licenses>
        <license>
            <name></name>
            <url></url>
        </license>
    </licenses>
    <developers>
        <developer>
            <id></id>
            <name></name>
            <email></email>
        </developer>
    </developers>
    <modules>
    </modules>
    <build>
        <plugins>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-compiler-plugin</artifactId>
                <version>3.11.0</version>
                <configuration>
                    <source>21</source>
                    <target>21</target>
                </configuration>
            </plugin>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-jar-plugin</artifactId>
                <version>3.3.0</version>
                <configuration>
                    <archive>
                        <manifest>
                            <mainClass>com.bootcamp.mavenassignment.App</mainClass>
                        </manifest>
                    </archive>
                </configuration>
            </plugin>
        </plugins>
    </build>
</project>

```

```

ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven_assignment$ mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.bootcamp.mavenassignment:maven_assignment >-----
[INFO] Building maven_assignment 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-jar-plugin/3.3.0/maven-jar-plugin-3.3.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-jar-plugin/3.3.0/maven-jar-plugin-3.3.0.pom (6.8 kB at 18 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/37/maven-plugins-37.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/37/maven-plugins-37.pom (9.9 kB at 88 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/37/maven-parent-37.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/37/maven-parent-37.pom (46 kB at 563 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/27/apache-27.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/27/apache-27.pom (20 kB at 261 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-jar-plugin/3.3.0/maven-jar-plugin-3.3.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-jar-plugin/3.3.0/maven-jar-plugin-3.3.0.jar (27 kB at 238 kB/s)
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ maven_assignment ---
[INFO] Deleting /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven_assignment/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ maven_assignment ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven_assignment/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:compile (default-compile) @ maven_assignment ---
[INFO] Changes detected - recompiling the module! :source
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 21] to target/classes
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ maven_assignment ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven_assignment/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:testCompile (default-testCompile) @ maven_assignment ---
[INFO] Changes detected - recompiling the module! :dependency
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 21] to target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ maven_assignment ---
[INFO] Surefire report directory: /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven_assignment/target/surefire-reports

-----
T E S T S
-----
Running com.bootcamp.mavenassignment.AppTest

```

```

-----
T E S T S
-----
Running com.bootcamp.mavenassignment.AppTest
Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.01 sec

Results :

Tests run: 1, Failures: 0, Errors: 0, Skipped: 0

[INFO]
[INFO] --- maven-jar-plugin:3.3.0:jar (default-jar) @ maven_assignment ---
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-archiver/3.6.0/maven-archiver-3.6.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-archiver/3.6.0/maven-archiver-3.6.0.pom (3.9 kB at 34 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-io/3.4.0/plexus-io-3.4.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-io/3.4.0/plexus-io-3.4.0.pom (6.0 kB at 64 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/4.4.0/plexus-archiver-4.4.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/4.4.0/plexus-archiver-4.4.0.pom (6.3 kB at 60 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/3.6.0/maven-archiver-3.6.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/3.6.0/maven-archiver-3.6.0.jar (79 kB at 1.1 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-io/3.4.0/plexus-io-3.4.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-io/3.4.0/plexus-io-3.4.0.jar (267 kB at 1.2 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/4.4.0/plexus-archiver-4.4.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/4.4.0/plexus-archiver-4.4.0.jar (211 kB at 924 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-interpolation/1.26/plexus-interpolation-1.26.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-interpolation/1.26/plexus-interpolation-1.26.jar (1.0 MB at 3.5 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-io/3.4.0/plexus-io-3.4.0.jar (79 kB at 1.1 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.4.2/plexus-utils-3.4.2.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.4.2/plexus-utils-3.4.2.jar (267 kB at 1.2 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/4.4.0/plexus-archiver-4.4.0.jar (211 kB at 924 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/4.4.0/plexus-archiver-4.4.0.jar (211 kB at 924 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-archiver/3.6.0/maven-archiver-3.6.0.jar (26 kB at 12 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-interpolation/1.26/plexus-interpolation-1.26.jar (85 kB at 38 kB/s)
[INFO] Building jar: /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven_assignment/target/maven_assignment-1.0-SNAPSHOT.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 7.038 s
[INFO] Finished at: 2026-02-06T21:49:12+05:30
[INFO] -----

```

```

ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven_assignment$ cd target/
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven_assignment/target$ ls
classes generated-sources generated-test-sources maven-archiver maven_assignment-1.0-SNAPSHOT.jar maven-status surefire-reports test-classes
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven_assignment/target$ java -jar maven_assignment-1.0-SNAPSHOT.jar
Hello World!
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven_assignment/target$

```

Q5) Differentiate between the different dependency scopes: compile, runtime, test, provided using different dependencies being defined in your pom.xml.

Compile scope

This is for libraries that my application needs all the time while writing code, running the program, and even when creating the final JAR.

Runtime scope

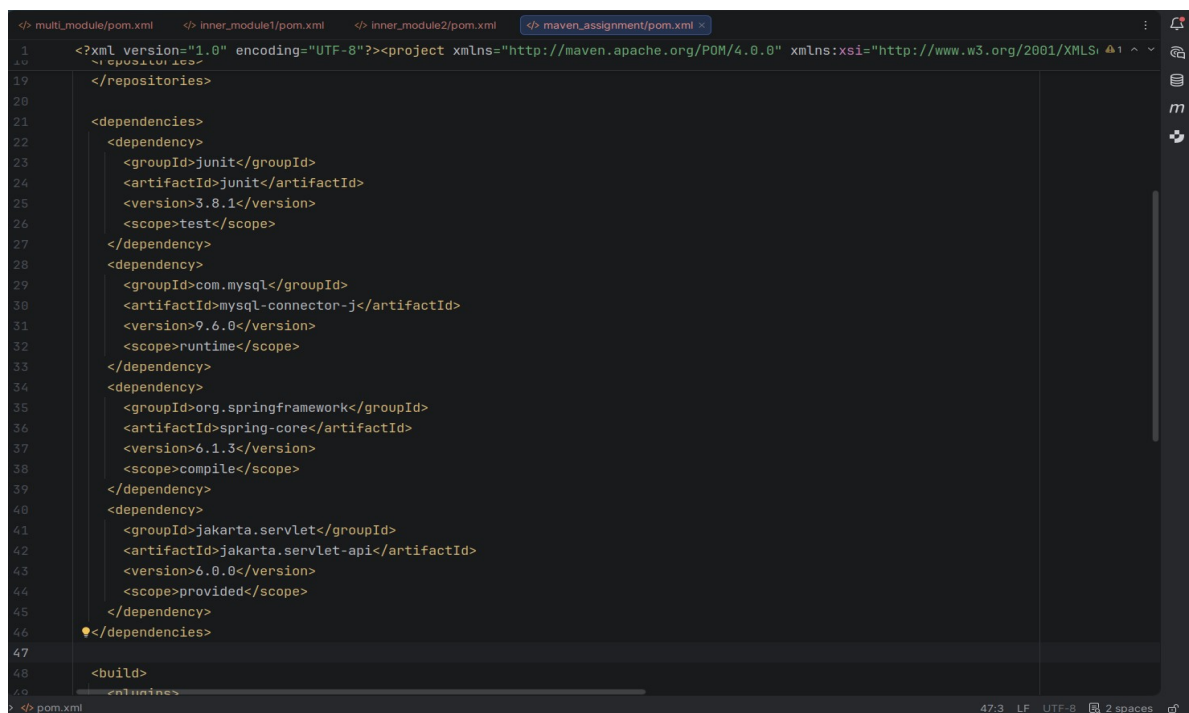
These dependencies are not required to compile the code, but are needed when the application **actually runs**.

Test scope

Used only for writing and running tests. These libraries are not included in the final JAR.

Provided scope

These dependencies are needed while compiling, but are already available in the runtime environment so Maven does not package them.



```
1  <?xml version="1.0" encoding="UTF-8"?><project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
2    <modelVersion>4.0.0</modelVersion>
3    <groupId>com.example</groupId>
4    <artifactId>maven-assignment</artifactId>
5    <version>1.0</version>
6    <packaging>jar</packaging>
7    <name>maven-assignment</name>
8    <description>Maven assignment</description>
9    <url>http://maven.apache.org</url>
10   <inceptionYear>2017</inceptionYear>
11   <repositories>
12     <repository>
13       <id>central</id>
14       <url>http://central.maven.org/maven2</url>
15     </repository>
16   </repositories>
17   <dependencies>
18     <dependency>
19       <groupId>junit</groupId>
20       <artifactId>junit</artifactId>
21       <version>3.8.1</version>
22       <scope>test</scope>
23     </dependency>
24     <dependency>
25       <groupId>com.mysql</groupId>
26       <artifactId>mysql-connector-j</artifactId>
27       <version>9.6.0</version>
28       <scope>runtime</scope>
29     </dependency>
30     <dependency>
31       <groupId>org.springframework</groupId>
32       <artifactId>spring-core</artifactId>
33       <version>6.1.3</version>
34       <scope>compile</scope>
35     </dependency>
36     <dependency>
37       <groupId>jakarta.servlet</groupId>
38       <artifactId>jakarta.servlet-api</artifactId>
39       <version>6.0.0</version>
40       <scope>provided</scope>
41     </dependency>
42   </dependencies>
43   <build>
44     <plugins>
45       <plugin>
46         <groupId>org.springframework.boot</groupId>
47         <artifactId>spring-boot-maven-plugin</artifactId>
48       </plugin>
49     </plugins>
50   </build>
51 </project>
```


Q6) Create a multi-module project. Run package command at the top level to make jar of every module.

```
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven$ mvn archetype:generate -DgroupId=com.bootcamp.mutlinmodule -DartifactId=multi_module -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] Parameter: basedir, Value: /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven
[INFO] Parameter: package, Value: com.bootcamp.mutlinmodule
[INFO] Parameter: groupId, Value: com.bootcamp.mutlinmodule
[INFO] Parameter: artifactId, Value: multi_module
[INFO] Parameter: packageName, Value: com.bootcamp.mutlinmodule
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven/multi_module
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.982 s
[INFO] Finished at: 2026-02-06T22:37:21+05:30
[INFO]
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven$ cd multi_module/
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven/multi_module$ ls
pom.xml  src
```

```
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven/multi_module$ mvn archetype:generate -DgroupId=com.bootcamp.innermodule1 -DartifactId=inner_module1 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.bootcamp.mutlinmodule:multi_module >-----
[INFO] Building multi_module 1.0-SNAPSHOT
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ multi_module >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ multi_module <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ multi_module ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] Parameter: basedir, Value: /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven/multi_module
[INFO] Parameter: package, Value: com.bootcamp.innermodule1
[INFO] Parameter: groupId, Value: com.bootcamp.innermodule1
[INFO] Parameter: artifactId, Value: inner_module1
[INFO] Parameter: packageName, Value: com.bootcamp.innermodule1
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven/multi_module/inner_module1
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.039 s
[INFO] Finished at: 2026-02-06T22:39:33+05:30
[INFO]
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven/multi_module$ mvn archetype:generate -DgroupId=com.bootcamp.innermodule2 -DartifactId=inner_module2 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] Reactor Build Order:
[INFO]
[INFO] multi_module [pom]
[INFO] inner_module1 [jar]
[INFO]
[INFO] -----< com.bootcamp.mutlinmodule:multi_module >-----
[INFO] Building multi_module 1.0-SNAPSHOT [1/2]
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ multi_module >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ multi_module <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ multi_module ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] Parameter: basedir, Value: /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven/multi_module
[INFO] Parameter: package, Value: com.bootcamp.innermodule2
[INFO] Parameter: groupId, Value: com.bootcamp.innermodule2
[INFO] Parameter: artifactId, Value: inner_module2
[INFO] Parameter: packageName, Value: com.bootcamp.innermodule2
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven/multi_module/inner_module2
[INFO]
[INFO] Reactor Summary for multi_module 1.0-SNAPSHOT:
[INFO]
[INFO] multi_module ..... SUCCESS [ 1.029 s]
[INFO] inner_module1 ..... SKIPPED
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.311 s
[INFO] Finished at: 2026-02-06T22:39:46+05:30
[INFO]
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven/multi_module$ ls
inner_module1 inner_module2 pom.xml  src
```

```
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven/multi_module$ mvn archetype:generate -DgroupId=com.bootcamp.innermodule2 -DartifactId=inner_module2 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] Reactor Build Order:
[INFO]
[INFO] multi_module [pom]
[INFO] inner_module1 [jar]
[INFO]
[INFO] -----< com.bootcamp.mutlinmodule:multi_module >-----
[INFO] Building multi_module 1.0-SNAPSHOT [1/2]
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ multi_module >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ multi_module <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ multi_module ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] Parameter: basedir, Value: /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven/multi_module
[INFO] Parameter: package, Value: com.bootcamp.innermodule2
[INFO] Parameter: groupId, Value: com.bootcamp.innermodule2
[INFO] Parameter: artifactId, Value: inner_module2
[INFO] Parameter: packageName, Value: com.bootcamp.innermodule2
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven/multi_module/inner_module2
[INFO]
[INFO] Reactor Summary for multi_module 1.0-SNAPSHOT:
[INFO]
[INFO] multi_module ..... SUCCESS [ 1.029 s]
[INFO] inner_module1 ..... SKIPPED
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.311 s
[INFO] Finished at: 2026-02-06T22:39:46+05:30
[INFO]
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven/multi_module$ ls
inner_module1 inner_module2 pom.xml  src
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven/multi_module$
```

```
<?xml version="1.0"?>
<project xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd" xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
    <modelVersion>4.0.0</modelVersion>

    <parent>
        <groupId>com.bootcamp.mutlimodule</groupId>
        <artifactId>multi_module</artifactId>
        <version>1.0-SNAPSHOT</version>
    </parent>

    <groupId>com.bootcamp.innermodule1</groupId>
    <artifactId>inner_module1</artifactId>
    <version>1.0-SNAPSHOT</version>
    <name>inner_module1</name>
    <url>http://maven.apache.org</url>

    <dependencies>
        <dependency>
            <groupId>junit</groupId>
            <artifactId>junit</artifactId>
            <version>3.8.1</version>
            <scope>test</scope>
        </dependency>
    </dependencies>
</project>
```

```
<?xml version="1.0"?>
<project xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd" xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
    <modelVersion>4.0.0</modelVersion>

    <parent>
        <groupId>com.bootcamp.mutlimodule</groupId>
        <artifactId>multi_module</artifactId>
        <version>1.0-SNAPSHOT</version>
    </parent>

    <groupId>com.bootcamp.innermodule2</groupId>
    <artifactId>inner_module2</artifactId>
    <version>1.0-SNAPSHOT</version>
    <name>inner_module2</name>
    <url>http://maven.apache.org</url>

    <dependencies>
        <dependency>
            <groupId>junit</groupId>
            <artifactId>junit</artifactId>
            <version>3.8.1</version>
            <scope>test</scope>
        </dependency>
    </dependencies>
</project>
```

```
<?xml version="1.0" encoding="UTF-8"?><project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema"
<modelVersion>4.0.0</modelVersion>
<groupId>com.bootcamp.mutlimodule</groupId>
<artifactId>multi_module</artifactId>

<packaging>pom</packaging>

<version>1.0-SNAPSHOT</version>
<name>multi_module</name>
<url>http://maven.apache.org</url>

<dependencies>
<dependency>
<groupId>junit</groupId>
<artifactId>junit</artifactId>
<version>3.8.1</version>
<scope>test</scope>
</dependency>
</dependencies>

<modules>
<module>inner_module1</module>
<module>inner_module2</module>
</modules>

<build>
<plugins>
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.11.0</version>
<configuration>
```

```
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven/multi_module$ mvn clean package
[INFO] Scanning for projects...
[INFO] -----
[INFO] Reactor Build Order:
[INFO]
[INFO] multi_module [pom]
[INFO] inner_module1 [jar]
[INFO] inner_module2 [jar]
[INFO]
[INFO] -----< com.bootcamp.mutlimodule:multi_module >-----
[INFO] Building multi_module 1.0-SNAPSHOT [1/3]
[INFO] -----[ pom ]-----
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ multi_module ---
[INFO]
[INFO] -----< com.bootcamp.innermodule1:inner_module1 >-----
[INFO] Building inner_module1 1.0-SNAPSHOT [2/3]
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ inner_module1 ---
[INFO] Deleting /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven/multi_module/inner_module1/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ inner_module1 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven/multi_module/inner_module1/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:compile (default-compile) @ inner_module1 ---
[INFO] Changes detected - recompiling the module! :source
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 21] to target/classes
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ inner_module1 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven/multi_module/inner_module1/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:testCompile (default-testCompile) @ inner_module1 ---
[INFO] Changes detected - recompiling the module! :dependency
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 21] to target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ inner_module1 ---
[INFO] Surefire report directory: /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven/multi_module/inner_module1/target/surefire-reports

-----
T E S T S
```



```
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ inner_module2 ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources
[INFO] skip non existing resourceDirectory /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven/multi_module/inner_module2/src/test/resources
[INFO] --- maven-compiler-plugin:3.11.0:testCompile (default-testCompile) @ inner_module2 ---
[INFO] Changes detected - recompiling the module! :dependency
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 21] to target/test-classes
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ inner_module2 ---
[INFO] Surefire report directory: /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven/multi_module/inner_module2/target/surefire-reports

-----
T E S T S
-----
Running com.bootcamp.innermodule2.AppTest
Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.013 sec

Results :

Tests run: 1, Failures: 0, Errors: 0, Skipped: 0

[INFO] --- maven-jar-plugin:2.4:jar (default-jar) @ inner_module2 ---
[INFO] Building jar: /home/ishmeet-singh/IdeaProjects/BootcampExercises/maven/multi_module/inner_module2/target/inner_module2-1.0-SNAPSHOT.jar
[INFO] -----
[INFO] Reactor Summary for multi_module 1.0-SNAPSHOT:
[INFO]
[INFO] multi_module ..... SUCCESS [ 0.088 s]
[INFO] inner_module1 ..... SUCCESS [ 1.414 s]
[INFO] inner_module2 ..... SUCCESS [ 0.335 s]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.915 s
[INFO] Finished at: 2026-02-06T22:45:28+05:30
[INFO] -----
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven/multi_module$ cd inner_module1/target/
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven/multi_module/inner_module1/target$ ls
classes generated-sources generated-test-sources inner_module1-1.0-SNAPSHOT.jar maven-archiver maven-status surefire-reports test-classes
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven/multi_module/inner_module1/target$ cd ../IdeaProjects/BootcampExercises/maven/multi_module/inner_module2/target
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven/multi_module/inner_module2/target$ ls
classes generated-sources generated-test-sources inner_module2-1.0-SNAPSHOT.jar maven-archiver maven-status surefire-reports test-classes
ishmeet-singh@TTNPL-ishmeet:~/IdeaProjects/BootcampExercises/maven/multi_module/inner_module2/target$
```